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Next item →

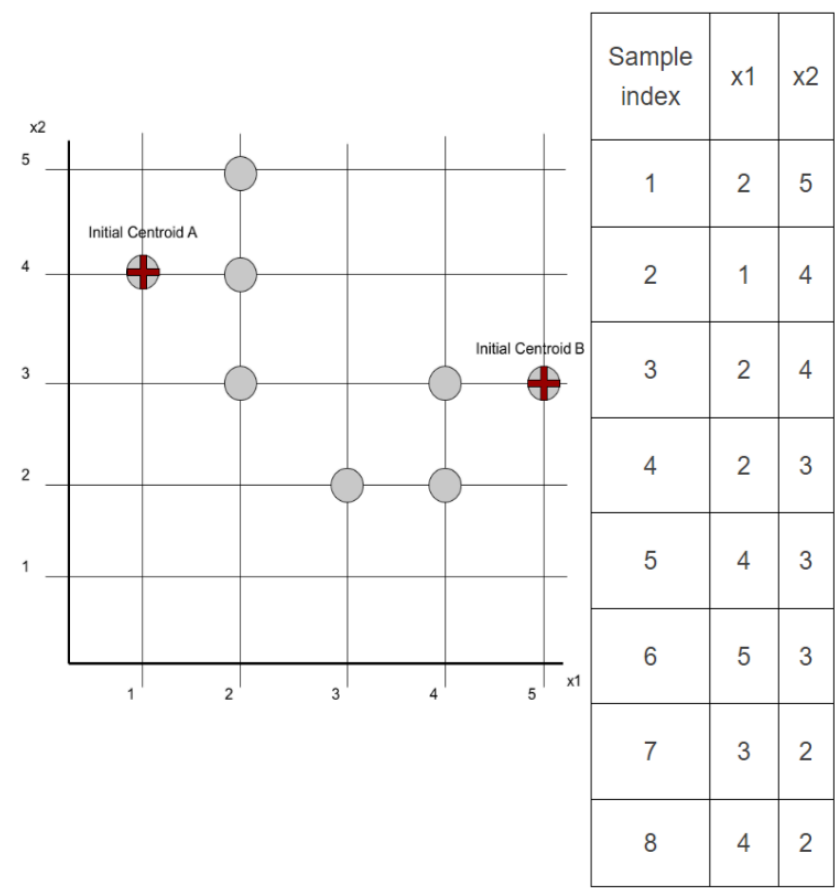
Questions 1 - 3

Consider the following initialization of a k-means clustering problem ($k = 2$ with clusters $\{A, B\}$). The raw data and a visualization of the space are provided below.

Note:

- We have samples indexed from 1 to 8 (sample_1, ..., sample_8)
- Centroid A is initialized at (1, 4); Centroid B is initialized at (5, 3)
- Use the Euclidean distance as your distance measure:

$$d(\text{sample}, \text{centroid}) = \sqrt{((x1_{\text{sample}} - x1_{\text{centroid}})^2 + (x2_{\text{sample}} - x2_{\text{centroid}})^2)}$$



1. What is the position of Centroid A after the first round of k-means optimization?

1 / 1 point

1 x1 = 1.75
2 x2 = 4

Nice

Good job

2. What is the p

1 x1 = 4
2 x2 = 2.5

Nice

Good job

3. What is the v means)?

Hint: the obj centroid. Its

objective fu

13

Nice work

Questions 4 - 7

You are given 10 points in a two-dimensional space as follows:

Point	X	Y
P1	5	8
P2	2	4
P3	0	0
P4	5	6
P5	3	3
P6	2	1
P7	9	8
P8	3	6
P9	5	1
P10	9	3

You want to cluster these points into three groups using the k-means algorithm. You can make use of the Euclidean distance formula:

$$d(\text{sample}, \text{centroid}) = \sqrt{((x1_{\text{sample}} - x1_{\text{centroid}})^2 + (x2_{\text{sample}} - x2_{\text{centroid}})^2)}$$

Select P2, P5, and P7 as the initial cluster centers. Perform iterations of the k-means algorithm and report the final cluster centers and the cluster assignments for each point.

4. What is the p

1 x1 = 1
2 y1 = 8.5

Nice

Good job

5. What is the p

1 x2 = 3.6
2 x2 = 4

Nice

Good job

6. What is the p

1 x3 = 7.67
2 y3 = 6.33

Nice

Good job

7. What is the c

1 p1 = 3
2 p2 = 2
3 p3 = 1
4 p4 = 2
5 p5 = 2
6 p6 = 1
7 p7 = 3
8 p8 = 2
9 p9 = 2
10 p10 = 3

Nice

Good job